Test 7

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February 20, 2025

- **Please submit a pdf with your answers on blackboard by Thursday 27th of February at 13:00.**

  Use the handouts to reply to these questions. Do not use generative AI tools. Note that each answer is in the handouts.

  You cannot answer using screenshots. You need to reply by typing or in your own writing. This promotes subconscious retention.

  This is an individual test. You cannot collaborate with other students or any other individual. This must be your work.

  You should be able to complete this in 1 page. Do not write, nor spend too much time on this.

Please state name, surname and CID.

1. (2 marks) Explain the two drawbacks of classical convergence analysis.
2. (2 marks) Define self-concordance for a function $f : \mathbb{R} \to \mathbb{R}$ and explain why it is coordinate invariant.
3. (2 marks) Consider a convex optimisation problem with equality constraints. Explain the method of reduction to unconstrained problems by elimination of the equality constraints.
4. (2 marks) Consider a convex optimisation problem with equality constraints. Explain the solution method based on the dual problem.
5. (2 marks) Explain why the infeasible start Newton's method is not a descent method.

Non-assessed questions, for self-study - Do not submit

- Provide an upper bound on the number of iterations until $f(x^{(k)}) - p^* \leq \varepsilon$ using self-concordance.
- Write the KKT system for Newton's method with equality constraints.
- Write the KKT system for the infeasible start Newton's method.
- Solve system

$$\begin{bmatrix} H & A^\top \\ A & 0 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix} = - \begin{bmatrix} g \\ h \end{bmatrix}$$

  using block elimination (assume that $H \succ 0$).